



University
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W5-04: Hadoop Distributed File System (Basics)

COMPSCI4064 (H) and COMPSCI5088 (M)

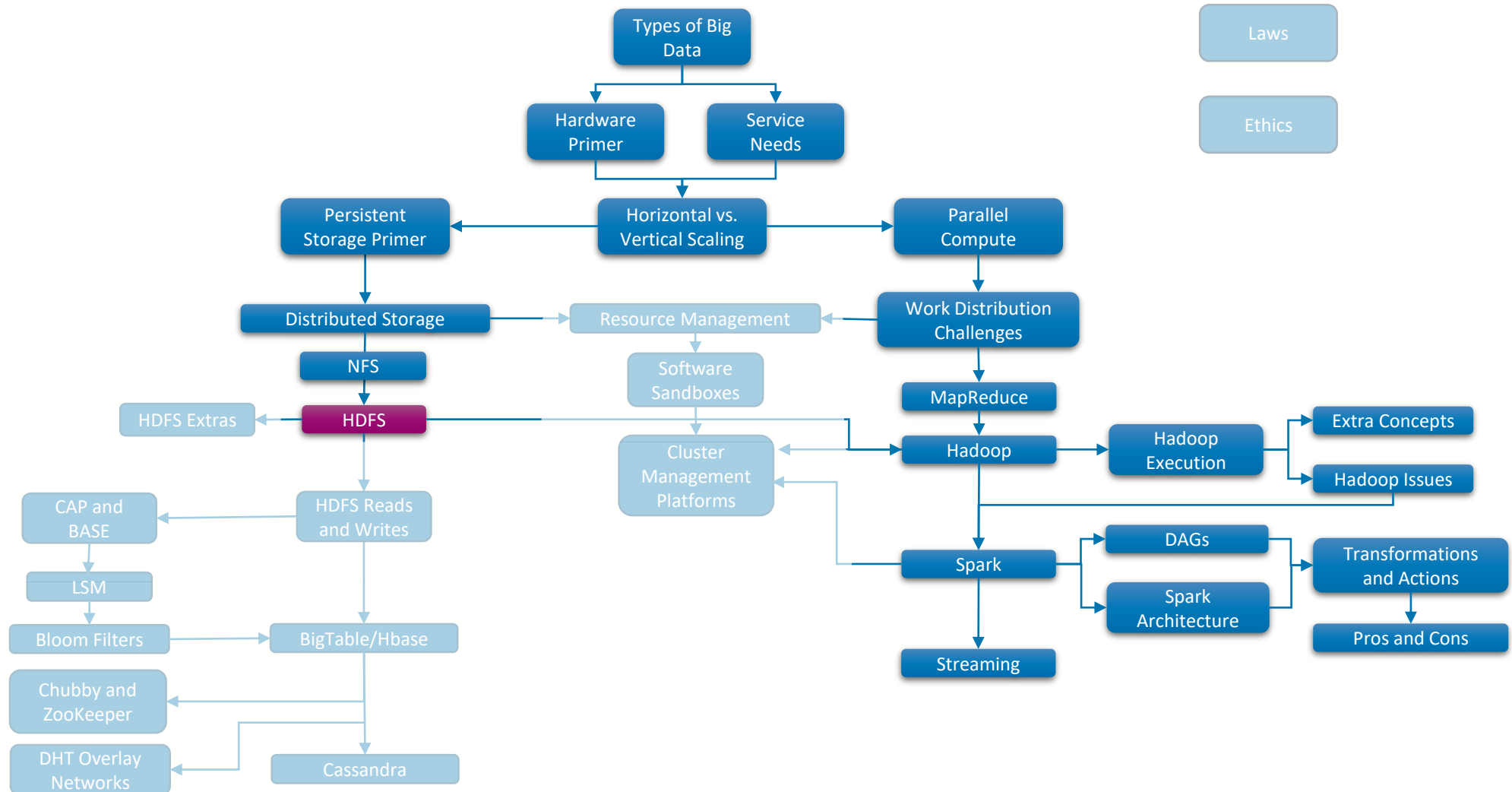
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Where Are We?





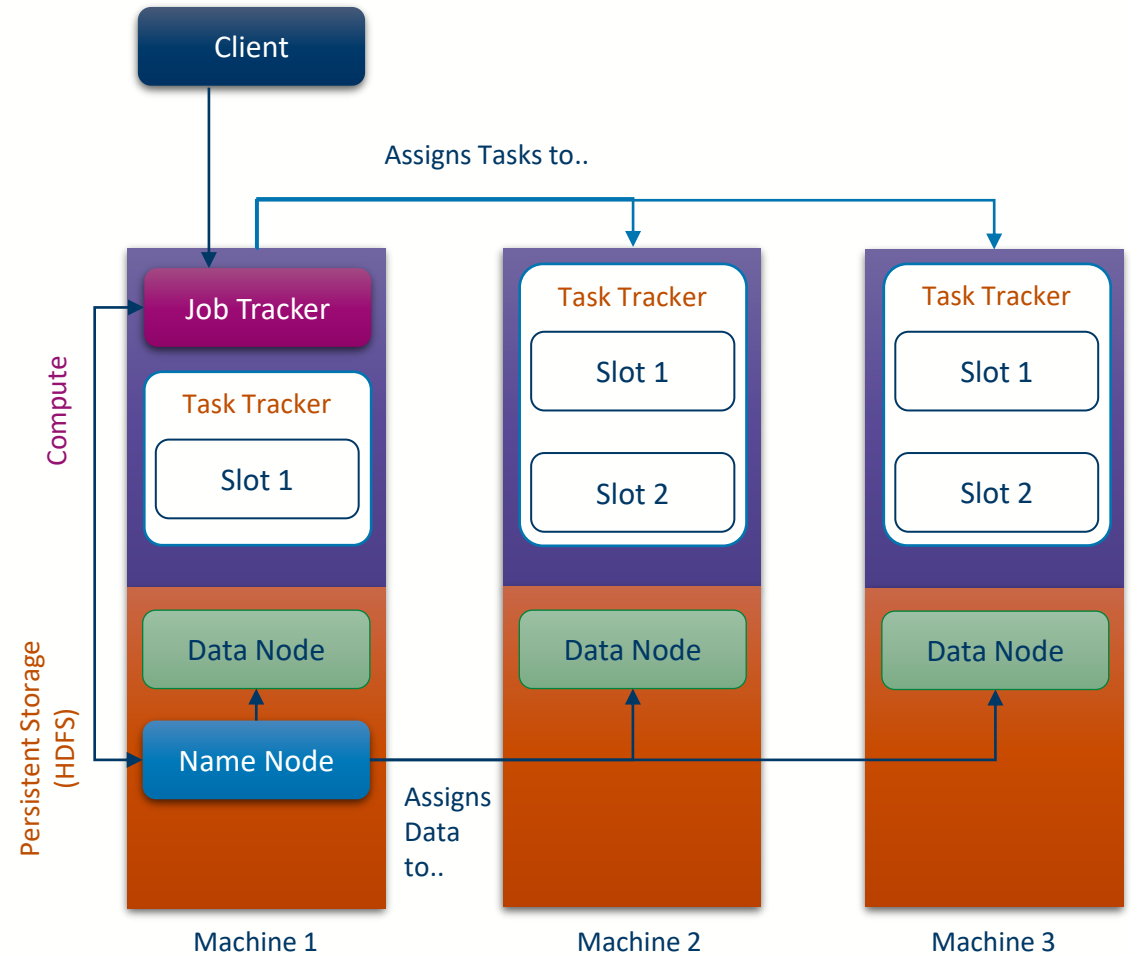
Hadoop Distributed File System (HDFS)

- The distributed file system (DFS) developed for the Hadoop implementation of MapReduce
- Designed to operate over a cluster of commodity machines
 - Off-the-shelf -- i.e., inexpensive but unreliable
- Designed for write-once-read-many access
 - High throughput batch storage and processing
 - High availability/fault tolerance
 - Scalability/elasticity
- All network communication over TCP/IP



Recall the Hadoop Architecture...

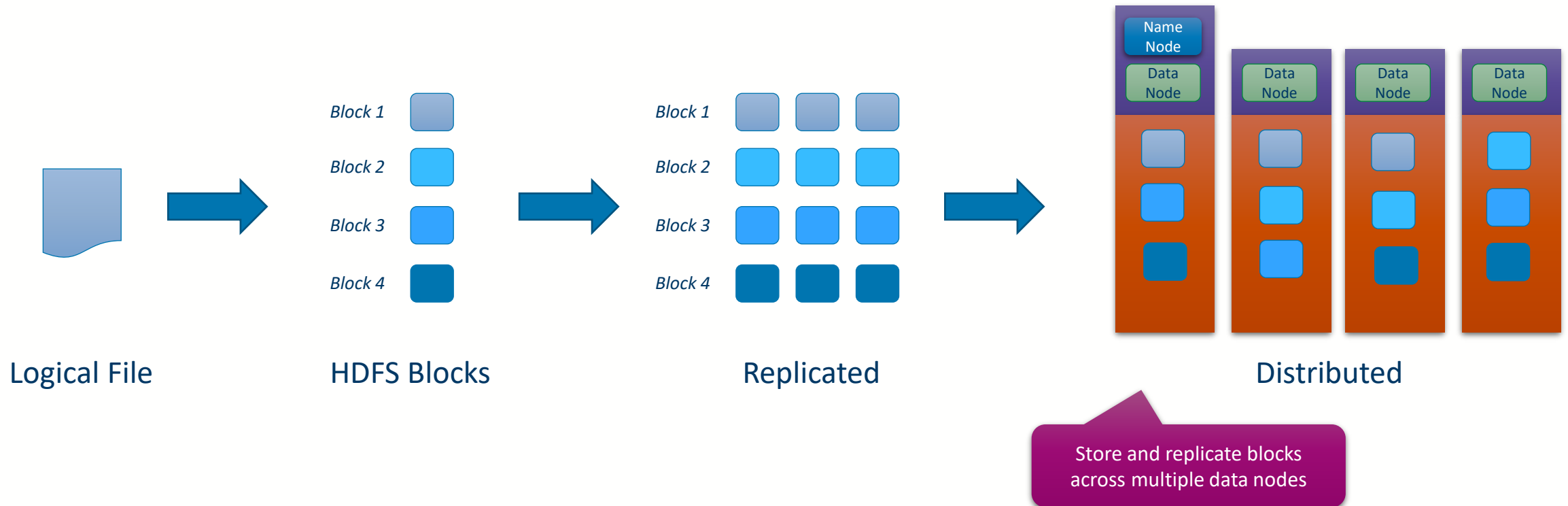
- Hadoop **Compute**
 - **Client**: Software that the user uses to submit work to the Hadoop cluster, communicates with the Job Tracker
 - **Job Tracker**: Receives work from the client, uses the Name Node to identify where data is located, assigns work to task trackers
 - **Task Tracker**: Executes Map and Reduce tasks
- Hadoop **HDFS**
 - **Name Node**: Tracks the location of all data held on the HDFS, handles replication
 - **Data Node**: Handles storage and access of the local files on the machine





HDFS Files

- The HDFS is a file system, so from the user perspective it stores files they provide
 - In practice an HDFS file is divided into one or more 'blocks'
 - Blocks are replicated amongst machines





Architectural Principles

- Look like a standard Unix file system, provides:
 - UNIX-like File System API for open/close, read/write, etc.
 - Unix-like index nodes (inodes)
 - Describes the metadata for a file-system object such as a file or a directory, e.g. ownership, access mode (read, write, execute permissions), and file type
 - Unix-like file paths, e.g. /my_directory/cool_file.txt
- Separates file content data from metadata
 - Namenode (NN): stores 'global' metadata (information about the files)
 - DataNodes (DN): stores file blocks and per-block metadata



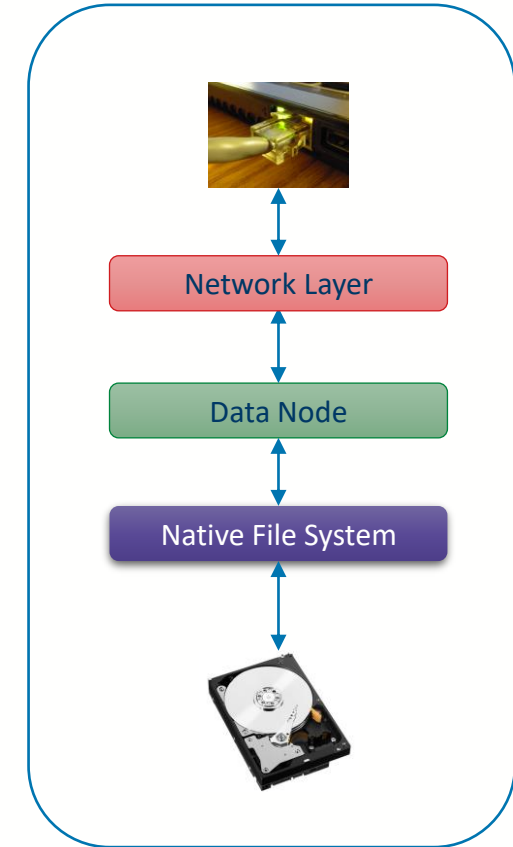
Name Node / Master Node

- The Name Node is responsible for **tracking files**, **managing the file system** and **holds the metadata** of all of the **stored files** within it.
 - Most importantly it holds the **number of blocks**, **locations** of the **data nodes** where those blocks are stored, and where the **replicas of those blocks are stored**
 - Also holds other critical metadata
 - The **state of the namespace**, i.e. the set of directories and their files and their hierarchical organization, along with a mapping of path names to unique file IDs
 - **Access permissions** for files
 - **Modification times** for files
- All this metadata is collectively called the filesystem's **image**
 - Is **loaded into RAM** for as long as the Name Node is running to keep access latency down (since most Hadoop services need information about the files/blocks)
 - Maintains **two recovery files**
 - A **snapshot** is stored locally at the Name Node in a so called checkpoint file in case it dies, enabling name node recovery
 - Also holds a journal file -- a **write-ahead log** (WAL) for all mutations to the filesystem since the last checkpoint – again allows for state recovery to the point of failure
- Note the name node is a **single point of failure** (until Hadoop v3)



Data Node

- Data Nodes are responsible for **storing each block** they are given, associated **read and write operations**, and **holding the block metadata**
- An **HDFS file block** is actually **two files** stored on the physical node's **native file system** (note this means that host FS will impact performance)
 - One file for the **contents of the HDFS file block**
 - One file for the **“local” metadata** (per block), i.e. Checksum and Size
- **Each block is BIG**: default = 128MBs
 - This is not like a normal file system
 - Big blocks are more compatible with the Big Data files and HDD storage the HDFS was designed for





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